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12 September 2026

Machine Learning in healthcare

Summary :

Linear regression and K-Nearest Neighbors (KNN) belong to basic machine learning algorithms, which have significant application in the domain of healthcare in addition to their standard applications. This paper highlights an application of linear regression in the process of monitoring of metastatic colorectal cancer and specifically, its role in analysis of changes in the biomarker of oxidative stress d-ROMs prior and post chemotherapy treatment. Through construction of a ratio of the d-ROM post-treatment level to pre-treatment level, linear regression allows distinguishing patients with the disease progression and those who respond well to the treatment. Similarly, in parallel with this, the paper reviews an unconventional application of KNN algorithm which uses the method not for classification purposes but for data imputation in the context of wearables health-monitoring technologies. Sensor dropout in the context of monitoring of geriatric frailty using smart bracelets leads to missing values, which KNN imputes by means of the nearest observation analogy. Thus, both these applications showcase how versatile relatively simple machine learning models can be — one provides clinically useful predictions about the disease progression while another serves as a fundamental data pre-processing approach in healthcare analytics.

Introduction :

Although machine learning algorithms are often explained using elementary examples in education, they do find relevance in the real world where they help in making decisions that influence patient monitoring, treatment evaluation, and even diagnosing patients. In this paper, we discuss two of such applications where linear regression is used for the analysis of the response of patients to colorectal cancer treatment, while K-Nearest Neighbors (KNN) algorithm is used to fill gaps in health monitoring records.

Medical terms explained :

Metastatic — describes cancer that has spread beyond its original site to other parts of the body. Metastatic colorectal cancer is a more advanced, harder-to-treat stage of the disease.

d-ROM (derivatives of reactive oxygen metabolites) — a specific biomarker that measures oxidative stress in the body (a byproduct of cellular damage, often elevated in cancer). Rising d-ROM levels have been linked to disease progression, while falling levels have been linked to a positive treatment response.

How linear regression predicts (colorectal cancer)

Collect the biomarker measurements — the measurement of the d-ROM level is taken from a patient's blood before and after chemotherapy treatment..

Compute a ratio — the model looks at the ratio of post-treatment to pre-treatment d-ROM levels for each patient.

Fit a linear relationship — through linear regression, one tries to find the best-fitting linear relationship between this d-ROM ratio and the actual clinical outcome of the patient, progression or response, given patient historical data.

Predict for new patients — once the model has learned this relationship, a new patient's d-ROM ratio can be plugged in to predict whether their cancer is likely progressing or responding to treatment — giving doctors an early, quantitative signal without waiting for imaging or more invasive tests.

KNN Imputation for Missing Wearable Data

The gap appears — There comes a time when a smart bracelet that is being used to monitor the physical weaknesses of an old person (in terms of movement, heart rate, and activity levels) doesn't record any data because of a technical problem.

KNN looks for "similar" data points — rather than making a blind guess, the algorithm will search for values among the other recorded values (of the same patient, and/or other patients) which match the most to the surrounding pattern; they are the "K nearest neighbors" to the missing value.

"Closeness" is based on overall pattern similarity — Two readings are said to be close if the other readings obtained from that same patient at the time (or readings of some other patients) resemble each other. For instance, if readings taken during another time of movement and heartbeat readings are very similar, then those are a possible choice for estimation.

The missing value is filled in using those neighbors — KNN essentially says: "this missing value probably looked similar to what we see in the closest matching cases," and uses those nearby values (often averaged) to make a sensible estimate, rather than leaving a blank or using a crude method like just plugging in the dataset's overall average.

The completed dataset supports downstream monitoring — once the gaps are filled, the health-monitoring system can reliably continue tracking the patient's frailty trend over time, without missing values distorting alerts or trend analysis that clinicians and caregivers rely on.

References

1. Miyamoto, Y., et al. (2023). Monitoring Metastatic Colorectal Cancer Progression According to Reactive Oxygen Metabolite Derivative Levels. *PMC*. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10705800/>
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